

TOTEM



USECASE BIBLE

THE VERIFIABLE OPERATING FABRIC
FOR THE AUTONOMOUS ECONOMY

A field guide for builders, investors, OEMs, infrastructure operators, communities and institutions exploring what becomes possible when identity, evidence, policy, physical action and programmable settlement share one stack.

SENSE > PROVE > DECIDE > ACT > SETTLE

01

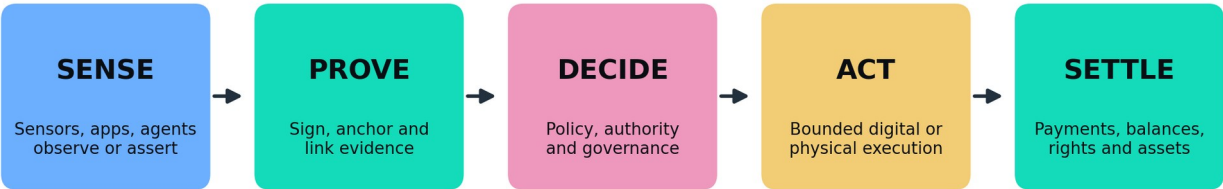
How to read this Bible

Totem is a fabric, not a single vertical.

The central idea. Totem Edge can be used wherever an observable or asserted fact needs to become trustworthy evidence, evidence needs to be evaluated against policy, policy needs to authorise an action, and rights or value need to move as a consequence. That applies to a DEX and a robot, a DAO and a cold-chain shipment, a digital product passport and a machine-insurance market.

A useful mental model

Facts → Rights → Decisions → Actions → Value. Totem’s differentiator is not any one arrow. It is that the same fabric can connect all five.



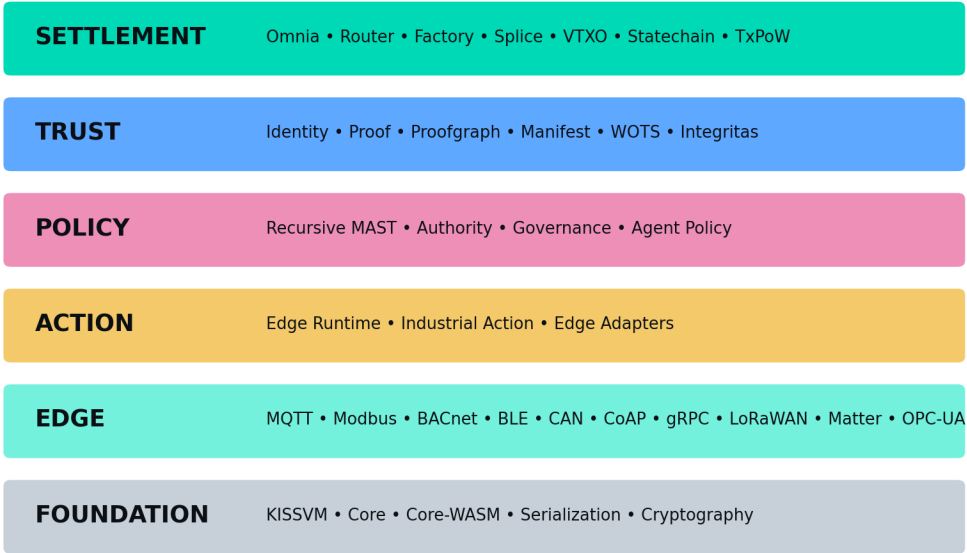
Every stage can produce evidence consumed by the next.

EVIDENCE LEGEND	
SHIPPED PRIMITIVE	A package/API exists in the SDK.
COMPOSABLE APPLICATION	Built by combining existing primitives.
EXPERIMENTAL TEMPLATE	A Recursive MAST/KISSVM template exists but is explicitly experimental / not audited.
FRONTIER CONCEPT	A plausible new market or product enabled by the composition of existing primitives; it still requires product, legal, risk and integration design.

02

The Totem operating fabric

One stack spanning digital apps, autonomous agents and the physical edge.



Shared sovereignty plumbing: Axia hosted access • self-hosted nodes • browser/PWA wallets • realtime • lookup • streams • pubsub • MCP

Delivery surfaces. Builders can present the same underlying fabric through conventional web apps, a self-custodial browser wallet, an installable PWA wallet, backend/server integrations, autonomous agents, hosted Axia chain access, self-hosted nodes, or edge runtimes on devices and gateways.

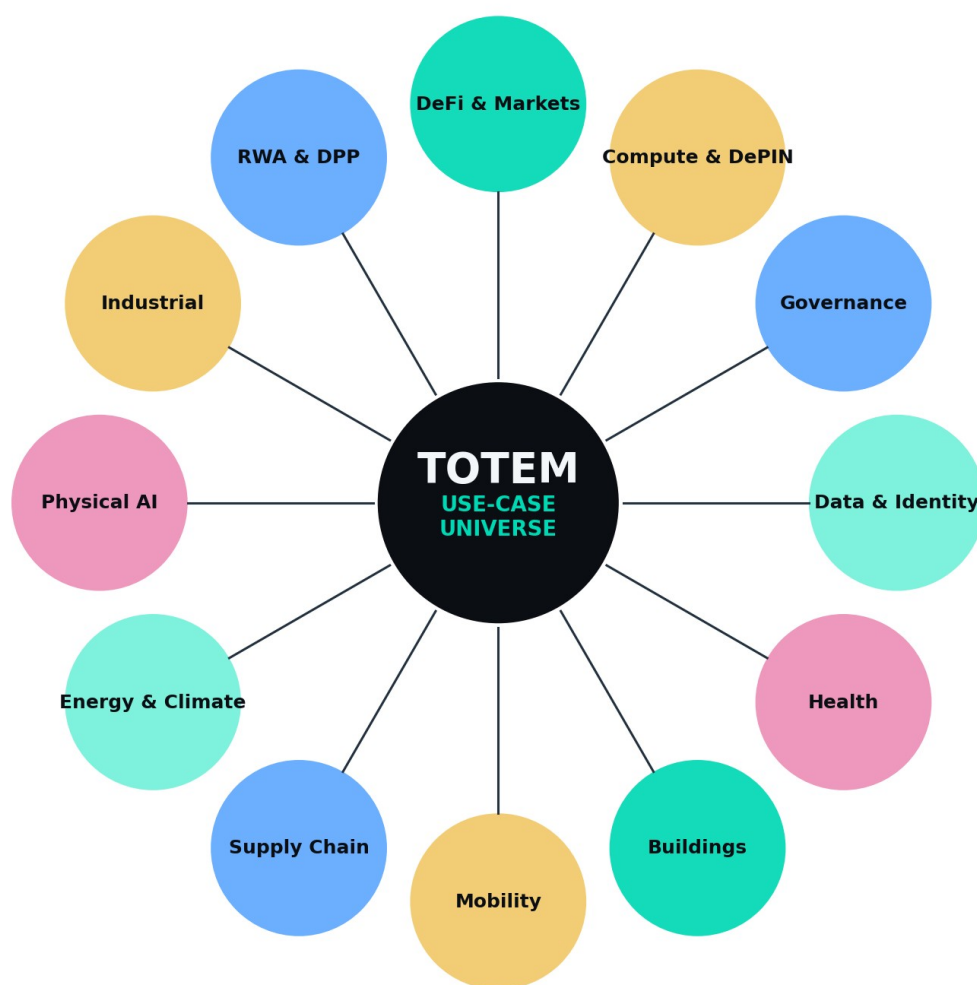
Surface	What it contributes	Best fit
Axia hosted access	Hosted RPC/chain access, broadcast, live portfolio streams and WOTS lease coordination.	Fastest path for consumer/web experiences and managed infrastructure.
Totem Browser Extension	Self-custodial browser signer and dApp provider surface.	DEXs, RWAs, DAOs, consumer apps, web-based machine control.
Totem PWA Wallet	Installable, cross-origin embeddable self-custodial wallet.	Mobile-first, field use, OEM-branded workflows, app-like experiences.
Server / MCP / agent surfaces	Server-side wallet, automation and agent-facing tooling.	Bots, enterprises, autonomous services, AI agents.
Edge / Pear / self-hosted	Transport-agnostic edge runtime, P2P and self-hosted infrastructure.	Factories, vehicles, farms, buildings, robots, sovereign deployments.

Why breadth matters

A conventional Web3 application can use the Browser Extension and Axia and never touch an industrial protocol. A machine can use edge adapters, proof and Omnia and never present a browser UI. The

03 Use-case atlas

The major verticals Totem can address today or enable through composition.



Markets & programmable finance — Payments, DEXs, swaps, prediction markets, launchpads, treasuries, Omnia-native state, machine finance.

Assets & lifecycle — RWA, digital product passports, certification, warranty, servicing, legal/title, circular economy.

Physical systems — Industrial, robotics, energy, buildings, mobility, agriculture, logistics, healthcare.

Information & trust — Identity, credentials, privacy, data markets, geospatial proof, digital twins.

Collective coordination — DAOs, cooperatives, Network States, public sector, shared infrastructure.

Autonomous infrastructure — Compute, telecoms, DePIN, machine-to-machine markets, AI agents.

Do not pigeon-hole the stack

The strongest opportunities often sit between categories: an RWA whose revenue is measured by the asset itself; a prediction market settled from edge telemetry; machine insurance bought for 30 minutes; a Network State that actually governs energy and compute assets.

04 Digital finance & programmable markets

The familiar Web3 surface - with a path into the physical economy.

Payments & digital money

Build ordinary payment products without giving up the ability to evolve into policy or machine-native commerce.

What builders can create	Totem composition	Likely buyers / operators
P2P and merchant payments • QR/payment requests • subscriptions and metered billing • escrow and conditional payments • payroll/revenue share • agent-initiated payments	connect + Browser/PWA + server + Axia + tx-builder/txpow + Omnia	Consumers, merchants, platforms, communities, OEMs

Frontier angle: Turn any verified physical event into a payment trigger.

DEX, swaps & capital markets

Totem exposes the transaction, wallet, channel, HTLC and routing surfaces needed to build trading experiences.

What builders can create	Totem composition	Likely buyers / operators
token swaps • routed and cross-token swaps • atomic HTLC exchange • OTC / RFQ markets • trading terminals • treasury rebalancing • liquidity and provider bond markets	connect + Omnia + router + HTLC + liquidity/provider bonds + wallet	Web3 builders, liquidity providers, treasuries

Frontier angle: Bridge asset markets to live physical evidence and machine-generated revenue.

Launchpads & token formation

Issue and coordinate communities or projects using wallet, token, policy and governance primitives.

What builders can create	Totem composition	Likely buyers / operators
token launches • community sales • crowdfunding • membership tokens • utility tokens • revenue-share structures • grants/public-goods funding	wallet + tx-builder + governance + treasury + identity/manifest	Projects, communities, DAOs, OEM ecosystems

Frontier angle: Launch an asset/service token that later becomes a control or revenue right over physical infrastructure.

Prediction & information markets

Markets can settle from human reporting, digital events or signed physical telemetry.

What builders can create	Totem composition	Likely buyers / operators
event markets • asset-price markets •	wallet + KISSVM + Omnia HTLC/asset +	Market operators, insurers, energy

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energy-generation forecasts • machine
uptime/failure markets • delivery arrival
markets • weather/crop markets •
charger availability markets

proof + edge/location/raster evidence

traders, logistics networks

Frontier angle: Edge-native markets where the oracle is a network of signed physical evidence rather than a single API.

05 Omnia-native economic systems

Beyond “payments”: programmable off-chain state for high-frequency relationships.

Micropayment networks

- pay-per-request APIs
- content streaming
- per-byte connectivity
- per-second compute
- per-reading data
- per-kWh energy

Conditional finance

- HTLC routing
- atomic swaps
- escrow
- delayed release
- service completion settlement

Treasury & membership

- governed spending caps
- member roots
- dividend pools
- cooperative revenue distribution

Asset state

- holder balances
- conservation rules
- off-chain asset transfer
- high-frequency bilateral accounting

Scale & operations

- multi-hop routing
- factories
- virtual channels
- splicing
- VTXO/offline patterns
- durable Omnia hosts

Builder insight

Omnia is best understood as a programmable economic state machine. The obvious use is payments; the differentiated use is to bind price, usage, membership, treasury or asset state to a long-lived relationship.

06

Real-world assets & Digital Product

Passports

Make the digital representation evolve with the physical asset.

A DIGITAL PRODUCT PASSPORT CAN BECOME AN OPERATING + ECONOMIC IDENTITY



identity • ownership • certification • maintenance • firmware • telemetry • rights • payments • residual value

RWA tokenisation & lifecycle

The SDK contains explicit RWA policy templates spanning tokenisation, fractionalisation, audits, income distribution, transfer, redemption and disposal.

What builders can create	Totem composition	Likely buyers / operators
real estate • infrastructure • equipment • commodities • receivables • IP/royalties • share transfer • rental/interest/dividend distribution	Recursive MAST RWA templates + identity + proof + asset/treasury programs + wallet	Asset issuers, funds, property/infrastructure owners, fintechs

Frontier angle: Build “living RWAs” where operating hours, production, maintenance and revenue are independently evidenced by the asset.

Digital Product Passports

A DPP can be more than a document repository: it can become the product’s operating and economic identity.

What builders can create	Totem composition	Likely buyers / operators
manufacture/serial identity • certification • ownership/custody • maintenance and calibration • firmware and software hashes • warranty • resale • recycling/end-of-life	Device lifecycle + supply-chain + proofgraph + identity + edge sensing + policy	OEMs, regulators, marketplaces, repair networks

Frontier angle: Use DPP state to decide whether a product may operate, be serviced, receive firmware, transfer ownership or access a market.

Certification & compliance-as-code

Turn multi-party certification into reusable policy roots and verifiable pipelines.

What builders can create	Totem composition	Likely buyers / operators
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product certification • operator credentials • supplier approval • permit-to-work • KYC/AML gates • environmental certification • cross-border recognition • audit trails	Compliance pipelines + recursive MAST + authority + identity/proof	OEMs, regulators, auditors, enterprises
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Frontier angle: A certification can become an executable permission - not just a PDF certificate.

07 Industrial systems, robotics & Physical AI

Where proofs, policy and settlement meet real actuation.

Industrial automation & smart manufacturing

Modbus, OPC-UA, MQTT, CAN, ROS2 and gRPC connect PLCs, control systems, robot cells and service planes to a common proof/policy runtime.

What builders can create	Totem composition	Likely buyers / operators
factory telemetry • condition monitoring • quality gates • proof-of-production • predictive maintenance • machine-as-a-service • pay-per-cycle • industrial data markets • guardrailed control	edge-* + edge runtime + proof + authority + industrial-action + Omnia	Manufacturers, system integrators, factories, industrial OEMs

Frontier angle: A financial or governance event can safely cause a bounded physical action - with an auditable receipt.

Robotics & Physical AI

Treat a robot as an identity with capabilities, a budget, policies, service manifests and verifiable actions.

What builders can create	Totem composition	Likely buyers / operators
delivery robots • warehouse AMRs • robot arms • inspection robots • agricultural robots • drones • autonomous construction/mining equipment • robot-as-a-service	ROS2/CAN/BLE + edge + identity + agent-policy + authority + industrial-action + Omnia	Robot OEMs, fleet operators, logistics, factories

Frontier angle: AI proposes; policy evaluates; governed keys sign; the robot acts; the result is proved and settled.

Machine-as-a-Service & servitisation

Commercial templates explicitly cover metered usage, feature licensing, warranty-conditioned servicing, escrow and leasing.

What builders can create	Totem composition	Likely buyers / operators
pay-per-cycle • pay-per-hour • pay-per-km • capacity subscriptions • feature unlocks • equipment leasing • outcome pricing • maintenance bundled into usage	Commercial MAST templates + meter program + edge sensing + Omnia	Industrial OEMs, equipment financiers, service companies

Frontier angle: Move OEM economics from one-off capex sales toward verifiable recurring machine revenue.

08

Energy, climate & the built environment

Meters and buildings become transacting, governed infrastructure.

Energy & utilities

The energy template family already models REC trading, microgrids, P2P trading, demand response and net metering.

What builders can create	Totem composition	Likely buyers / operators
smart metering • prosumer markets • microgrids • virtual power plants • battery arbitrage • demand response • EV charging • REC issuance/retirement • grid services • water/gas metering	Energy templates + meter + Modbus/BACnet/MQTT + Omnia + governance	Utilities, energy communities, DER operators, charger networks

Frontier angle: Machines can negotiate and settle energy at the edge while communities govern price and policy.

Climate markets & MRV

Edge evidence can strengthen measurement/reporting/verification for environmental instruments.

What builders can create	Totem composition	Likely buyers / operators
renewable certificates • carbon/soil carbon MRV • energy-efficiency credits • biodiversity evidence • water-quality incentives • recycling credits	edge sensing + proof/integritas + location/spatial/raster + policy + asset tokens	Climate projects, registries, corporates, auditors

Frontier angle: Link credit issuance to verifiable device- or imagery-derived evidence instead of manual reporting alone.

Smart buildings & hospitality

BACnet, Matter and BLE bring buildings, rooms and shared facilities into the same identity/policy/payment model.

What builders can create	Totem composition	Likely buyers / operators
HVAC/lighting • smart locks • occupancy • energy performance • tenant utilities • parking/EV • workspace booking • hotel credentials • facility maintenance	BACnet/Matter/BLE + authority + statechain + Omnia + DPP	Building owners, proptech, hotels, facilities managers

Frontier angle: A property can expose services, access rights and energy flexibility as machine-readable markets.

09

Supply chains, mobility & field systems

Bring evidence and settlement to environments that are mobile, intermittent and multi-party.

Supply chain & logistics

The SDK includes explicit provenance, cold-chain, bill-of-lading, customs, inventory and quality-inspection policy patterns.

What builders can create	Totem composition	Likely buyers / operators
provenance • chain of custody • cold-chain • proof of pickup/delivery • bills of lading • customs • inventory • quality gates • milestone settlement • trade finance	Supply-chain MAST + BLE/LoRaWAN/Modbus + proofgraph + location + Omnia	Logistics firms, pharma/food, ports, customs tech, insurers

Frontier angle: One evidence chain can drive payment, financing, insurance and compliance at the same time.

Mobility & automotive

CAN and location proof support vehicles and heavy machinery as identities with measurable usage and controllable rights.

What builders can create	Totem composition	Likely buyers / operators
vehicle telemetry • vehicle DPP • mileage/usage proof • fleet management • car sharing • pay-per-mile • tolls/parking • EV charging • robotaxi/drone settlement	CAN/BLE + location/spatial + DPP + authority + wallet/Omnia	OEMs, fleets, insurers, mobility platforms

Frontier angle: Finance, insurance and access can be priced from the vehicle's own verifiable operating history.

Agriculture & environmental field systems

LoRaWAN and low-connectivity operation make the stack suitable for farms and remote sensing.

What builders can create	Totem composition	Likely buyers / operators
soil and weather sensing • precision irrigation • livestock tracking • crop traceability • water allocation • farm equipment leasing • crop insurance • carbon farming • yield markets	LoRaWAN + edge proof + location + policy + Omnia	Farms, agritech, co-ops, insurers, commodity networks

Frontier angle: The same field sensor can supply operations, financing, insurance and environmental markets.

10 Healthcare, data & identity

Sensitive systems need explicit consent, provenance and bounded authority.

Healthcare & medtech

Healthcare templates cover device regulation, patient consent, clinical trials, prescription verification and regulated data access.

What builders can create	Totem composition	Likely buyers / operators
medical-device identity • post-market surveillance • patient consent • clinical trial compliance • prescription validation • provider credentials • pharma provenance • remote monitoring	Healthcare MAST + identity + proof + BLE + privacy policy + wallet	Medtech OEMs, hospitals, research networks, pharma

Frontier angle: A medical device’s certification and operational evidence can travel with it across institutions and jurisdictions.

Data markets & digital twins

Telemetry can be licensed, sold per reading or used to feed a live digital twin with proof of provenance.

What builders can create	Totem composition	Likely buyers / operators
sensor data markets • industrial telemetry • vehicle/energy/building data • digital twin feeds • pay-per-reading • time/purpose-limited licences • AI-training data licensing	edge + proof + telemetry licence + privacy + Omnia meter	Data owners, OEMs, marketplaces, AI companies

Frontier angle: Turn operational data into a monetisable asset while preserving explicit policy over who may consume it.

Privacy & consent infrastructure

Recursive policy can make data access purpose-, category-, processor- and time-bounded.

What builders can create	Totem composition	Likely buyers / operators
consent receipts • revocation • retention rules • data subject requests • portability • conditional data release • ZK claim integration	Data-privacy templates + identity + proof + authority	Health, finance, mobility, enterprise data platforms

Frontier angle: Data-use rights become machine-verifiable and auditable instead of buried in application terms.

Identity, credentials & access

Humans, devices, agents and services can each carry identity and delegated authority.

What builders can create	Totem composition	Likely buyers / operators
dApp sign-in • device identity • employee/technician credentials • visitor access • role/capability access • time-bounded permissions • service identity • recovery/rotation	root-identity + identity + proof + authority + access-control templates	Enterprises, OEMs, dApps, facilities, institutions

Frontier angle: The same credential model can authenticate a website user and authorise a valve operation.

11 Compute, connectivity & DePIN

Autonomous services can discover, meter, price and settle infrastructure usage.

Compute & AI infrastructure

Usage-meter economics map naturally onto compute and AI services.

What builders can create	Totem composition	Likely buyers / operators
GPU/CPU markets • pay-per-inference • edge compute • serverless billing • storage/bandwidth bundles • capacity reservations • agent-procured compute • model/data licensing	edge/service manifests + meter + Omnia + provider bonds + agent policy	Data centres, GPU operators, AI platforms, marketplaces

Frontier angle: Agents can autonomously discover and procure compute under explicit wallet budgets and provider reputation.

Telecoms & connectivity

MachinePay’s canonical model is connectivity paid by usage.

What builders can create	Totem composition	Likely buyers / operators
paid Wi-Fi • community broadband • pay-per-byte • LoRaWAN gateway markets • roaming • backhaul sharing • satellite/ground station access • network SLA settlement	MQTT/stream/pubsub + manifest + meter + Omnia + provider bond	ISPs, community networks, IoT operators, DePIN builders

Frontier angle: Connectivity itself becomes a discoverable machine service rather than a human subscription product.

DePIN & shared infrastructure

Totem can support familiar DePIN categories but add identity, policy and physical action to reward/payment mechanics.

What builders can create	Totem composition	Likely buyers / operators
sensor networks • wireless • compute/storage • energy/charging • mapping • robot/drone networks • community infrastructure	edge + proof + manifest + lookup + bonds + Omnia + governance	DePIN projects, infrastructure co-ops, OEM networks

Frontier angle: Move from “token incentives for nodes” toward verifiable service markets with operational control.

12 Governance, Network States & institutional systems

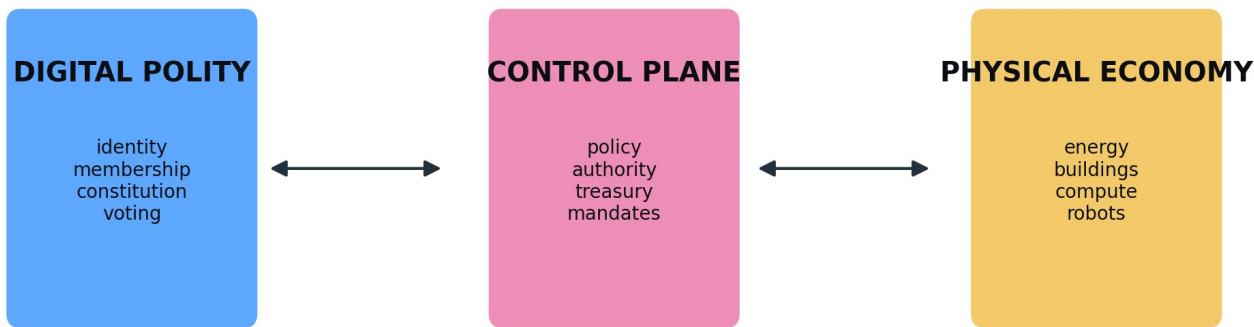
Collective decisions can become executable mandates over digital and physical resources.

DAOs & cooperatives

Governance primitives include linear, quadratic and liquid models plus proposal-to-mandate execution.

What builders can create	Totem composition	Likely buyers / operators
treasury governance • member admission/removal • grants • public goods • investment/RWA DAOs • energy co-ops • data co-ops • fleet governance	governance + authority + recursive MAST + treasury program + wallet	Communities, protocols, co-ops, asset owners

Frontier angle: A DAO can govern machines, sites and service policy - not only treasury transfers.



Network States can govern real productive infrastructure - not only tokens and forums.

Network States & digital polities

A Network State can combine membership, constitution, treasury and voting with ownership/control of real productive infrastructure.

What builders can create	Totem composition	Likely buyers / operators
citizenship/membership • local chapters • mutual recognition • public goods • procurement • community currency • land/RWA registries • energy/connectivity/compute ownership	identity + governance + recursive MAST + RWA + edge + Omnia	Online communities, charter communities, digital nations, co-ops

Frontier angle: The polity can govern energy, buildings, compute and robots through the same constitutional control plane.

Public sector & smart-city infrastructure

The stack can also be used without token-centric governance.

What builders can create

permits/licences • procurement • asset registries • participatory budgets • transport/parking • environmental sensing • utilities • public infrastructure maintenance

Totem composition

identity + authority + governance + edge + proof + payment rails

Likely buyers / operators

Municipalities, agencies, smart-city integrators

Frontier angle: Turn public rules into auditable policy while preserving human decision points and bounded machine execution.

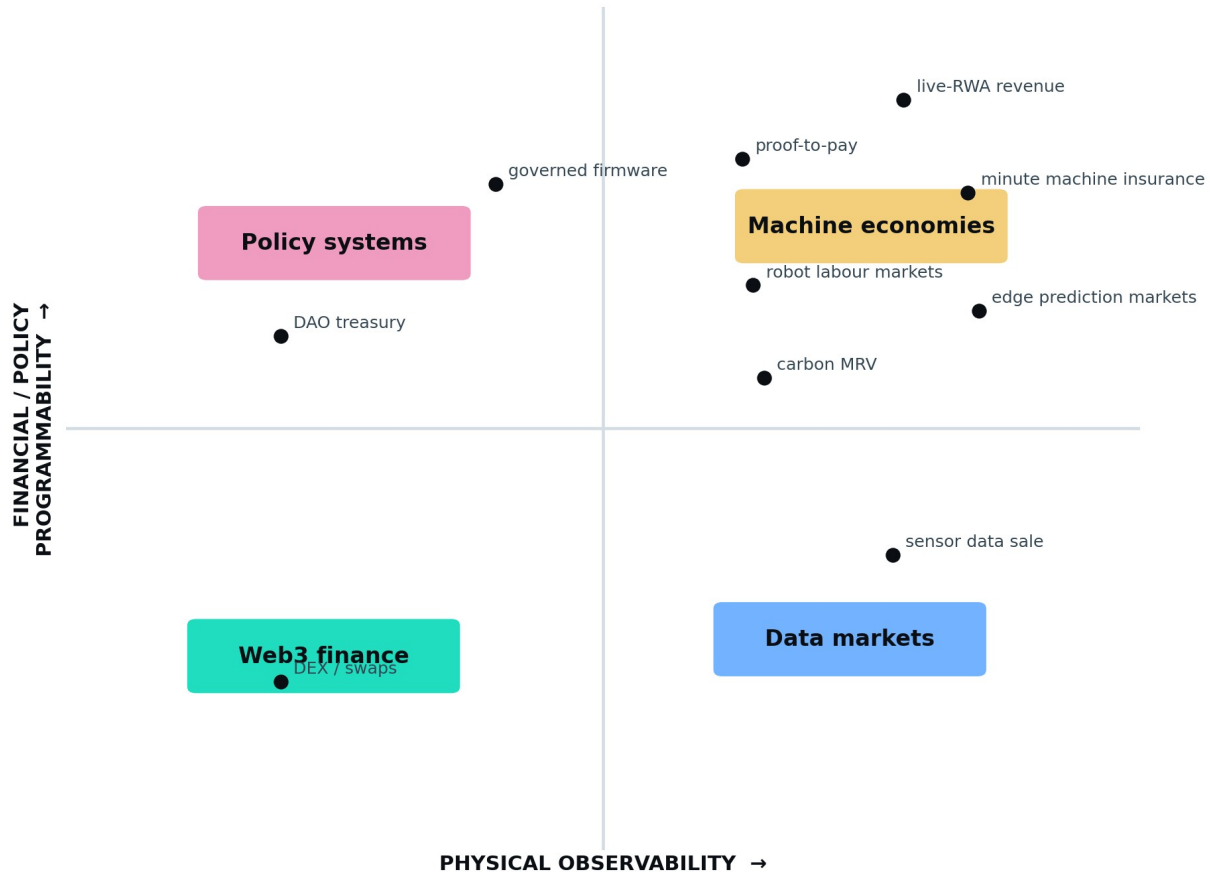
13 Additional verticals

The same primitives extend into other sectors with surprisingly little conceptual change.

Insurance & risk	Parametric, usage-based and event-specific cover; telematics; cargo/cold-chain; downtime; warranty insurance; peer mutuals; proof-triggered claims.
Legal & notarisation	Document notarisation, proof of existence, contracts, deeds, powers of attorney, witness flows, cross-jurisdiction recognition.
Warranty & aftermarket	Maintenance-conditioned warranties, approved parts/technicians, repair escrow, residual-value evidence, refurbishment and resale.
Retail & unattended commerce	Vending, kiosks, lockers, parking, car washes, pay-per-use appliances, offline-capable settlement.
Creator & IP markets	Patent/software/model/data licences, royalties, pay-per-use rights, IP-backed assets and revenue-share structures.
Gaming & interactive worlds	State channels, tournaments, prize escrow, asset ownership, guild treasuries, prediction/skill markets.
Education & workforce	Academic/professional credentials, training, operator certification, site permissions, credential revocation and mutual recognition.
Circular economy	Repair/reuse histories, battery second life, material passports, recycling incentives, deposit-return, producer responsibility.
Geospatial & remote sensing	Proof of location/route, geofenced rights, imagery provenance, land/forestry/agriculture evidence, disaster/insurance assessment.

14 Frontier crossovers

The products that emerge only when physical evidence, policy and programmable economics share one fabric.



01 Micro-duration machine insurance — A robot buys 30 minutes of breakdown/collision cover before entering a hazardous work zone. Live telemetry can influence eligibility, price and claims evidence.

02 Machine uptime prediction markets — Trade “will turbine 17 remain above 95% availability for six hours?” against signed machine telemetry.

03 Predictive-maintenance insurance — A machine’s condition evidence changes its cover terms or triggers mandatory service before continued operation.

04 Autonomous maintenance escrow — The machine detects a fault; an agent procures a technician; proof of successful repair releases escrow.

05 Live-revenue RWA — Tokenise a charger, robot, solar inverter, vending machine or GPU cluster and distribute independently evidenced operating revenue.

06 Proof-backed receivables — A verified delivery or machine job automatically becomes financing-grade evidence for a receivable.

07 Dynamic equipment leasing — Lease price, limits and residual value respond to operating hours, location, health and maintenance evidence.

08 Autonomous robot charging — Robots discover chargers, compare price/reputation, open channels, charge and settle without human interaction.

09 Edge energy prediction markets — Buildings, batteries or substations trade forecasts and settle against meter proofs.

- 10 Building flexibility markets** — HVAC systems bid load reduction into demand-response markets and prove the actual curtailment through BACnet telemetry.
- 11 Cold-chain finance** — The same signed temperature history influences delivery payment, invoice financing and cargo insurance.
- 12 Data-as-a-cash-flow asset** — Tokenise a sensor network; buyers pay per reading; revenue flows automatically to asset holders.
- 13 Consent-paid data** — A person permits a defined dataset for a defined purpose and period; every use creates an auditable receipt and micropayment.
- 14 Machine reputation bonds** — Service providers or machines post bonds alongside verifiable performance histories and service manifests.
- 15 Autonomous procurement** — A factory agent detects a spare-part need, validates approved supplier/part policy and budget, orders, pays and records the outcome.
- 16 Smart spare parts** — A machine refuses an unauthorised replacement part because the part's identity/passport cannot satisfy policy.
- 17 Pay-to-unlock capability** — A machine feature activates only while a valid entitlement, subscription or maintenance state is present.
- 18 Dynamic residual-value markets** — Equipment resale price is based on provable usage, servicing, firmware and operating history.
- 19 Physical capacity markets** — Trade future factory, GPU, battery, charging, warehouse or robot time as reservable economic capacity.
- 20 Robot labour markets** — Robots advertise signed manifests for delivery, inspection, welding or cleaning; buyers pay against completed-task proof.
- 21 Drone slot markets** — Location and route proof plus time/geography-bounded authority produce machine-verifiable flight rights.
- 22 Water allocation markets** — Meters establish actual consumption; policy defines entitlements; channels settle transfers.
- 23 Insurance-mutual DAOs of machines** — Asset owners fund a common treasury and settle objective claims using edge evidence.
- 24 Autonomous infrastructure cooperatives** — Community-owned energy, compute, chargers and network nodes earn revenue and are governed by their users.
- 25 Compliance-conditioned finance** — Loan, lease or insurance terms respond to independently verifiable maintenance, safety or environmental compliance.
- 26 Proof-to-pay** — Delivery, kWh, machine cycle, inspection, repair or task completion becomes a direct settlement trigger.
- 27 Proof-to-act** — Payment cleared → valve opens; credential verified → door opens; insurance active → machine starts.
- 28 Vote-to-act** — A passed proposal produces an authority mandate that changes how real machines or sites are allowed to behave.
- 29 Policy-to-market** — An asset automatically enters/exits markets depending on certification, owner, geography, health or regulatory state.
- 30 DPP with embedded economics** — One persistent product identity spans manufacture → certification → finance → operation → data revenue → warranty → resale → recycling.

15 Audience lenses

How different stakeholders should read the opportunity map.

Builders

Start with one surface and one hard problem. A DEX may need only Connect + Wallet + Omnia. A robot service may need Edge + Policy + Omnia. Use the fabric to expand later, not to make the first release complex.

OEMs

Think beyond telemetry dashboards. Device identity + DPP + policy + metering + settlement creates recurring-service, licensing, warranty and aftermarket business models.

Investors

The breadth is portfolio-like: digital finance provides familiar entry points, while edge/proof/policy create differentiation and new crossover markets. Evaluate which primitives are production-qualified versus experimental.

Infrastructure operators

Use service manifests, bonds, metering and channels to make energy, compute, connectivity or machine capacity directly purchasable by software agents and devices.

Enterprises

Policy/authority can provide an auditable control plane over both spending and physical operations, while keeping integration transport-agnostic.

Communities / DAOs / Network States

The stack can govern productive assets - energy, buildings, compute, connectivity, robots - not only tokens and online proposals.

Regulators / auditors

The strongest fit is not “regulation by blockchain”; it is independently replayable evidence, scoped authority, explicit policy roots and verifiable lifecycle records.

16 Reusable build patterns

A small number of recipes generate a very large number of products.

Web3 app

Web UI → @totemsdk/connect → Browser Extension/PWA → Axia or self-hosted chain access → transaction/Omnia.

Verified sensor market

Protocol adapter → Edge runtime → signed proof → policy gate → per-reading Omnia meter → buyer wallet.

Governed machine

Sensor/command adapter → policy tree → authority mandate → industrial-action guardrails → receipt → settlement.

Living RWA

Physical asset identity → DPP/proofgraph → telemetry proofs → RWA ownership/rights → revenue distribution.

Autonomous agent

AI proposal → agent-policy → authority → wallet/Omnia → receipt; optionally extend to physical action.

Network State infrastructure

Membership/identity → governance → mandate bridge → treasury → owned physical infrastructure → service revenue.

Parametric risk product

Evidence source → policy threshold → insured state → escrow/vault → proof-triggered claim/settlement.

Policy-controlled data

Data source → proof/provenance → consent/licence policy → metered access → payment → usage receipt.

The practical rule

Do not deploy every subsystem for every use case. Totem's value is composability. A minimal product should use the smallest trusted path that solves the problem, then add identity, policy, edge evidence or settlement modules as the business model requires.

17 Exhaustive inspiration index

A rapid-scan list for workshops, investor sessions and product ideation.

Finance & markets

P2P payments • merchant checkout • invoices • subscriptions • micropayments • escrow • timelocks • payroll • revenue share • DEX • atomic swaps • cross-token swaps • OTC/Rfq • liquidity markets • launchpads • crowdfunding • prediction markets • treasury management • agent trading • portfolio apps

Assets & lifecycle

real estate • infrastructure • equipment • commodities • receivables • IP • royalties • fractionalisation • audits • custody • dividends • redemption • DPP • warranty • maintenance • repair • resale • recycling • material passports

Industrial & Physical AI

PLCs • SCADA • robot cells • CNC • condition monitoring • quality control • proof-of-production • predictive maintenance • MaaS • pay-per-cycle • feature licensing • equipment leasing • robot labour • autonomous procurement • drones • AMRs • heavy machinery

Energy & climate

smart meters • P2P energy • microgrids • VPPs • battery arbitrage • demand response • net metering • EV charging • RECs • carbon MRV • energy efficiency • grid services • water/gas metering • community energy

Supply chain & mobility

provenance • custody • cold-chain • bill of lading • customs • inventory • quality inspection • proof of delivery • trade finance • vehicle telemetry • fleet management • car sharing • tolls • parking • pay-per-mile • mobility insurance • drone delivery

Buildings & hospitality

HVAC • lighting • smart locks • access control • occupancy • parking • EV charging • tenant utilities • workspace booking • hotel credentials • facility maintenance • HOA/co-op governance

Health & life sciences

medical-device identity • regulatory evidence • post-market surveillance • patient consent • clinical trials • prescriptions • provider credentials • health-data access • wearables • remote monitoring • pharma provenance • vaccine cold-chain

Data, identity & privacy

sensor data markets • telemetry licensing • digital twins • pay-per-reading • purpose-limited data • AI training data • consent • revocation • portability • ZK claims • human/device/agent identity • credentials • service identity • zero-trust access

Governance & institutions

DAOs • quadratic voting • liquid democracy • cooperatives • RWA DAOs • energy co-ops • fleet governance • Network States • local currencies • public goods • procurement • permits • smart cities • participatory budgeting • mutual recognition

Compute, telecoms & DePIN

GPU markets • pay-per-inference • edge compute • storage • bandwidth • Wi-Fi markets • community broadband • LoRaWAN gateways • roaming • satellite access • SLA settlement • sensor networks • energy DePIN • robot networks

Legal, education, creator & circular

notarisation • proof of existence • contracts • powers of attorney • professional credentials • training records • operator certification • patent/software/model licences • royalties • gaming state channels • guild treasuries • repair/reuse • battery second life • deposit return • recycling incentives

18 Responsible positioning

Inspiration should be ambitious without pretending every application is already production-qualified.

- Treat SDK primitives, packages and wallet surfaces as engineering building blocks - not guarantees that a vertical is legally or operationally ready.
- Recursive MAST domain templates are explicitly experimental/not audited; use them as design accelerators and policy examples, then perform independent review before production deployment.
- Physical control, medical, financial, insurance, public-sector and regulated RWA systems require domain-specific safety, compliance, licensing and integration work beyond the SDK.
- Prediction markets and parametric risk products need explicit oracle, manipulation, dispute and jurisdiction design even when the evidence layer is cryptographically strong.
- Autonomous agents should remain bounded by policy and authority; the stack's architecture deliberately allows signing keys to remain outside the agent.
- For investor and partner discussions, distinguish “supported primitive”, “composable solution” and “frontier product concept” rather than presenting all ideas at the same maturity level.

The right ambition

Totem's breadth is real because the primitives compose across domains. Credibility comes from being precise about which layer is already implemented and which layer still needs product, integration or assurance work.

19 Source anchors

Primary repository material behind this map.

Totem SDK README: <https://github.com/Totem-Edge/totem-sdk/blob/main/README.md>

Totem Edge Runtime Grey Paper:

https://github.com/Totem-Edge/totem-sdk/blob/main/TOTEM_EDGE_GREY_PAPER.md

Governance & Authority Green Paper:

https://github.com/Totem-Edge/totem-sdk/blob/main/TOTEM_GOVERNANCE_GREEN_PAPER.md

Omnia README: <https://github.com/Totem-Edge/totem-sdk/blob/main/packages/omnia/README.md>

@totemsdk/connect README: <https://github.com/Totem-Edge/totem-sdk/blob/main/packages/connect/README.md>

Recursive MAST templates: <https://github.com/Totem-Edge/totem-sdk/tree/main/packages/recursive-mast/src/templates>

Totem Browser Extension: <https://github.com/Totem-Edge/totem-sdk/tree/main/extensions/totem-extension>

Totem PWA Wallet: <https://github.com/Totem-Edge/totem-sdk/tree/main/extensions/totem-pwa-wallet>

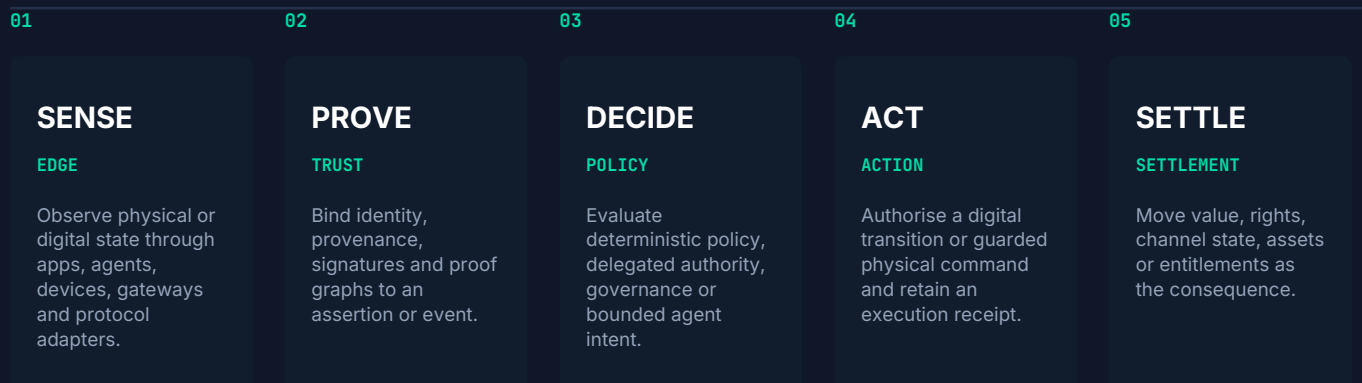
TOTEM EDGE

Facts. Rights. Decisions. Actions. Value.

Build the web app. Build the machine. Build the institution. Build the crossover.

One operating fabric

The architecture and delivery surfaces that make the full application universe coherent.



Delivery surfaces

WEB	@totemsdk/connect + Browser Extension Self-custodial browser signing and dApp access.
MOBILE	PWA Wallet Installable, cross-origin embeddable wallet workflows.
HOSTED	Axia RPC, broadcast, portfolio streams and WOTS lease coordination.
AGENTS	Server + MCP Backend automation and policy-bounded agent transactions.
PHYSICAL	Edge + Pear Device, gateway, peer and self-hosted execution.

FACTS > RIGHTS > DECISIONS > ACTIONS > VALUE

The 36-domain universe

Domains 01-09. Every required family is explicitly represented and traceable to its principal treatment.

01	Payments, wallets & digital money P2P, merchant checkout, invoices, subscriptions, escrow, payroll and human/machine/agent payments.	SEE 04
02	DEX, swaps & capital markets Routed swaps, HTLC exchange, RFQ/OTC, liquidity, trading, launchpads and token markets.	SEE 04
03	Prediction & information markets Events, prices, weather, logistics, energy and machine outcomes settled against defined evidence.	SEE 04
04	Omnia-native financial products Programmable payment, meter, HTLC, vault, treasury, membership and asset channel state.	SEE 05
05	Real-world assets Tokenisation through disposal, including telemetry-linked ownership, finance and cash flow.	SEE 06
06	Digital Product Passports Operational identity from manufacture and certification through service, resale and recycling.	SEE 06
07	Certification & compliance-as-code Credentials, issuer/revocation checks, permits, KYC/AML and multi-authority policy.	SEE 06
08	Supply chain, trade & logistics Provenance, custody, cold chain, customs, inventory, trade finance and proven delivery.	SEE 09
09	Industrial automation & manufacturing PLCs, SCADA, factories, condition monitoring, proof of production and bounded control.	SEE 07

The 36-domain universe

Domains 10-18. Every required family is explicitly represented and traceable to its principal treatment.

10	Robotics & Physical AI	SEE 07
	Policy-bounded AI proposals, governed signing, robot action receipts and fleet economics.	
11	Energy, utilities & climate	SEE 08
	Meters, microgrids, flexibility, EV charging, RECs, carbon MRV and utility settlement.	
12	Buildings, real estate & hospitality	SEE 08
	BACnet/Matter environments, access, facilities, utility billing, property and shared governance.	
13	Automotive, mobility & transport	SEE 09
	CAN telemetry, fleet identity, usage finance/insurance, charging, tolls, routes and slots.	
14	Agriculture, food & environment	SEE 09
	LoRaWAN sensing, irrigation, provenance, farm finance, parametric risk and conservation.	
15	Healthcare, medtech & life sciences	SEE 10
	Device identity, trials, consent, regulated access, wearables and pharmaceutical provenance.	
16	Telecoms & connectivity	SEE 11
	Pay-per-byte access, roaming, bandwidth and gateway markets, SLA settlement and discovery.	
17	Compute, AI infrastructure & data centres	SEE 11
	GPU/CPU markets, inference, storage, capacity, agent procurement and metered billing.	
18	Data markets, telemetry & digital twins	SEE 10
	Pay-per-reading/query, licensed feeds, data provenance, access receipts and AI datasets.	

The 36-domain universe

Domains 19-27. Every required family is explicitly represented and traceable to its principal treatment.

19	Geospatial, mapping & location	SEE 10/13
	Location/route proofs, geofencing, remote-sensing provenance, land and inspection evidence.	
20	Privacy, consent & personal data	SEE 10
	Purpose, category, processor and time-bounded access, revocation, portability and ZK claims.	
21	Identity, credentials & access	SEE 10
	Human, device, agent, organisation and service identity with delegated, rotatable authority.	
22	Device security & software lifecycle	SEE 06/07
	Commissioning, key rotation/recovery, revocation, governed firmware, anti-rollback and decommissioning.	
23	Warranty, servicing & aftermarket	SEE 13
	Usage-conditioned cover, approved parts/technicians, repair escrow, resale and residual value.	
24	Insurance & risk	SEE 13/14
	Parametric, usage/event-specific and micro-duration cover with proof-triggered claims.	
25	Legal, notarisation & instruments	SEE 13
	Existence proofs, timestamps, agreements, delegated signing, escrow and recognition.	
26	Governance, DAOs & cooperatives	SEE 12
	Treasuries, membership, delegation, proposals, mandates and governance-triggered action.	
27	Network States & digital polities	SEE 12
	Constitutional roots, citizenship, public goods, assets and physically operating services.	

The 36-domain universe

Domains 28-36. Every required family is explicitly represented and traceable to its principal treatment.

28	Public sector & smart cities	SEE 12
	Permits, procurement, infrastructure, utilities, public audit and bounded emergency authority.	
29	Retail, vending & unattended commerce	SEE 13
	Kiosks, lockers, rentals, gates, loyalty and offline-capable machine settlement.	
30	Machine-as-a-Service & servitisation	SEE 07
	Pay-per-cycle/task/hour, feature unlocks, equipment leases and outcome pricing.	
31	Creator, IP & licensing markets	SEE 13
	Software, models, datasets, patents and media rights with royalties and usage receipts.	
32	Gaming, interactive apps & communities	SEE 13
	State channels, tournaments, assets, guild treasuries, prediction and fan markets.	
33	Education, skills & workforce	SEE 13
	Portable credentials, training, qualifications, operator permissions and pay-on-completion.	
34	Circular economy & end of life	SEE 13
	Repair, reuse, second life, material passports, take-back, deposits and recycling evidence.	
35	Physical infrastructure / DePIN	SEE 11
	Sensing, wireless, compute, energy, mobility and robot networks with proof and governance.	
36	Autonomous-agent economy	SEE 02/11/16
	Delegated budgets, procurement, APIs, compute/data purchases, receipts and human escalation.	

Frontier crossovers: 31-40

Ten required crossover concepts completing the 30 already profiled in Section 14.

31 Usage-based warranties

Coverage persists only while operating-envelope, maintenance and approved-part evidence satisfies policy.

32 EV-charger prediction + reservation

Agents trade or reserve short future availability windows, then settle against charger evidence.

33 Autonomous microgrid economies

Machines buy and sell energy while residents or asset owners govern rules through a cooperative.

34 Autonomous cargo insurance

Leg-specific cover attaches at custody transfer and expires automatically at the next proven handoff.

35 Equipment-native carbon / REC issuance

Meter evidence from operating equipment becomes the issuance and retirement evidence layer.

36 Governed firmware

Manufacturer, regulator, owner and fleet policy jointly determine installable versions and anti-rollback rules.

37 Curb, road & parking markets

Vehicles acquire short-duration location rights that infrastructure can verify locally.

38 Waste & recycling markets

Passport-backed disposal releases deposits, refunds, producer obligations or sustainability incentives.

39 Network States with physical economies

A digital polity owns and governs energy, buildings, connectivity, compute and robots.

40 Safety-conditioned machine finance

Leasing, insurance or payment terms improve while certified operating-envelope compliance is proved.

Coverage audit

What the Bible covers, where it is strongest, and what requires a separate technical specification.

STRONG**Digital finance**

Payments, DEX/RFQ, swaps, launchpads, information markets and treasury patterns.

STRONG**Programmable state**

Omnia channels, HTLCs, routing, membership, asset state and durable hosts.

STRONG**Assets & lifecycle**

RWA lifecycle, DPPs, certification, warranty, service and circularity.

STRONG**Physical systems**

Industry, Physical AI, energy, buildings, mobility, logistics and agriculture.

STRONG**Trust & information**

Identity, proof, consent, data markets, spatial evidence and digital twins.

STRONG**Collective systems**

DAOs, cooperatives, Network States, public infrastructure and mandates.

STRONG**Autonomous infrastructure**

Compute, telecoms, DePIN, agents and machine-to-machine markets.

COVERED**Commercial audiences**

Builder, OEM, operator, institution, investor and regulator lenses.

COVERED**Composition patterns**

Seven reference patterns connect surfaces, primitives and outcomes.

COVERED**Responsible positioning**

Maturity labels, audit caveats, oracle risk, agent bounds and regulation.

What is deliberately not claimed

Coverage completeness does not mean production readiness.

1

Security and assurance

The Bible identifies shipped, composable, experimental and frontier layers, but it is not an audit, threat model, key-management standard or production security case.

2

Protocol guarantees

Latency, throughput, availability, privacy and cost depend on deployment choices. They must be benchmarked for each product rather than inferred from a use-case pattern.

3

Oracle and evidence quality

Signed evidence proves provenance and integrity, not necessarily truth. Sensor calibration, redundancy, fraud models, disputes and human override remain application responsibilities.

4

Regulatory design

RWA, payments, markets, insurance, health and public-sector deployments require jurisdiction-specific legal analysis, licences, consumer protections and governance.

5

Integration completeness

Adapters and reference patterns reduce integration work; they do not remove OEM certification, field testing, safety engineering, operational monitoring or support obligations.

6

Commercial validation

The atlas demonstrates addressable solution spaces. Buyer urgency, procurement cycles, pricing, unit economics and channel strategy still require vertical discovery.

THE CORRECT READING

This Bible is a possibility map and composition guide. Product requirements, architecture decisions, assurance evidence and launch criteria belong in the specification for each implementation.

Brand system used in this edition

Derived from the live Totem Browser identity at totem.ing.

Core assets

Official eagle mark and condensed TOTEM wordmark. The mark is used as a confident identifier, not decorative clip-art; the wordmark remains clear of competing headlines.

	#0F1729	Dark foundation
	#00DBA8	Primary signal
	#00D9B5	Secondary teal
	#FFFFFF	Primary type
	#94A3B8	Secondary type

Typography and layout

Inter leads the identity, coverage and navigation pages; JetBrains Mono handles labels, metadata and technical pathways. A restrained editorial serif remains on the dense field-guide pages for long-form reading. Dark foundations, bright signal accents, fine rules, generous spacing and modular cards extend the product interface into a professional publication system.

ONE FABRIC / MANY SURFACES / BOUNDED CLAIMS